Joshua Keene, PhD

(810) 287-5699 | keenejosh22@gmail.com | keenejosh.github.io | ORCID: 0009-0006-5104-2480

PROFESSIONAL SUMMARY

Neuroscience PhD specializing in neuroinflammation, tauopathy, and Alzheimer's disease pathology using transgenic and knock-in mouse models. Doctoral research demonstrated p-tau (AT8) elevations of 2.8-fold in hippocampus and 3.0-fold in the whole hemisphere 30 days after the final systemic PAMP challenge in PS19 mice; first-author manuscript submitted to ASN Neuro. Master's research demonstrated a 46% (male) and 59% (female) reduction in hippocampal amyloid plaque burden following prenatal inflammatory stress in 5xFAD mice. Research experience spans in vivo preclinical studies, primary cell culture and in vitro assays, quantitative histology, advanced imaging, and biostatistical analysis in R. Prior clinical neuropsychology experience provides direct exposure to neurocognitive profiles across TBI, dementia, and neurodevelopmental populations.

EDUCATION

2018–2024 **PhD in Neuroscience**

Central Michigan University, Mount Pleasant, MI | Advisor: Dr. Gary Dunbar

Dissertation: *PAMP's Exposure Increases the Areas of Pathological Tau and Inflammation in the PS19 Mouse Model of Alzheimer's Disease*.

2015–2018 **MS in Experimental Psychology**

Central Michigan University, Mount Pleasant, MI | Advisor: Dr. Yannick Marchalant

Master's Thesis: *Role of Early Exposure to Inflammatory Stress on Plaque Deposition and Microglia Density in Alzheimer's Disease Using the 5xFAD Mouse Model*.

2010–2013 **BA in Psychology, Minor in Philosophy**

Michigan State University, East Lansing, MI

RESEARCH INTERESTS

Neuroinflammation | Aging | Alzheimer's Disease | Neurodegeneration | Tauopathy | Amyloid Pathology | Innate Immune Signaling | Glial Biology | Peripheral Immune Activation | Preclinical Mouse Models

MANUSCRIPTS UNDER REVIEW

In Prep **Joshua Keene**, Taylor Desjarlais, Brianna Jackman, Swathi Suresh, Julien Rossignol, Marcia Gordon, Gary L. Dunbar, David Morgan. Sequential Systemic Challenge with Four Distinct PAMPs Causes Persistent Increases of Tau Phosphorylation and Glial Activation in PS19 Tauopathy Mice. Manuscript submitted for publication, ASN Neuro (2026).

JOURNAL PUBLICATIONS

2015 Rohling, M. L., Meyers, J. E., Williams, G. R., Kalat, S. S., Williams, S. K., & **Keene, J**. (2015). Application of the Daubert Standards to the Meyers Neuropsychological Battery Using the Rohling Interpretive Method. *Psychological Injury and Law, 8(3), 252-264*.

PROFESSIONAL EXPERIENCE

2022–2024 **Collaborative Graduate Researcher**

Laboratory of Dr. David Morgan and Dr. Marcia Gordon | Michigan State University, Grand Rapids Research Center (GRRC), Grand Rapids, MI

- Served as primary cross-institutional researcher (CMU/MSU-GRRC) on a multi-year preclinical study examining whether repeated systemic administration of multiple PAMPs (Zymosan A, Poly I:C, ODN 2395, LPS) drives tau pathology and neuroinflammation in PS19 tauopathy mice; first-author manuscript in final preparation for submission.
- Demonstrated p-tau (AT8) elevations of 2.8-fold in hippocampus and 3.0-fold in whole-hemisphere 30 days post-final PAMP challenge in PS19 mice, with total human tau (HT7) remaining unaltered, indicating increased tau phosphorylation at pS202/pT205 independent of changes in total human tau abundance, and supporting peripheral immune activation as a modifiable contributor to tau pathology.
- Identified positive Spearman correlations between p-tau burden and glial reactivity markers in the hippocampus (MHC II: rs=0.523; GFAP: rs=0.556; CD68: rs=0.501), supporting an association between p-tau accumulation and glial activation following peripheral immune challenges.
- Performed microtome sectioning and DAB IHC on hemi-brain tissue across a comprehensive antibody panel targeting markers of tau pathology (HT7, AT8) and neuroinflammation (GFAP, MHC II, CD68), with additional exploratory immunostaining for Iba1, CD45, CD11b, and CD11c without quantitative analysis; acquired whole-slide digital images via ZEISS Axio Scan.Z1 and quantified percent immunopositive area using NearCYTE with marker-specific standardized thresholds applied consistently across all experimental groups; conducted sandwich ELISA on fresh-dissected hippocampal and cortical tissue; scored and analyzed NOR and SOR behavioral data using ANY-maze software.
- Conducted parallel 3xTg cohort to test whether amyloid pathology modulates PAMP-induced tau pathology and neuroinflammation relative to the tau-driven PS19 model; observed absence of expected amyloid pathology through 6E10,

Congo Red, and H31L21 validation staining, revealing phenotypic divergence consistent with genetic drift and supporting discontinuation of the model line.

- Conducted all biostatistical analysis in R and GraphPad Prism, selecting non-parametric methods based on distributional and variance assessments; applied ART-ANOVA for genotype x treatment interactions, Mann-Whitney U tests for PS19-specific tau markers, and Spearman correlations for glial-tau relationships; reported bootstrapped effect sizes (Cliff's Delta, Hodges-Lehmann; 5,000-10,000 resamples) with Holm-corrected post-hoc contrasts.

2020–2024 **Doctoral Researcher**

Laboratory of Dr. Gary Dunbar | Central Michigan University, Mount Pleasant, MI

- Conducted doctoral dissertation research investigating whether repeated systemic PAMP administration affects tau pathology and neuroinflammation in PS19 tauopathy mice; independently managed all aspects of a 150+ animal study across PS19 and 3xTg cohorts including colony maintenance, breeding, and genotyping (tail biopsy, PCR); completed doctoral dissertation in 2024.
- Executed in vivo experimental procedures including PAMP preparation and dilutions, monthly IP injections over a four-month period, systematic post-injection monitoring of sickness behavior including hunched posture, hypoactivity, piloerection, with supportive care (mash diet) until recovery, and behavioral testing (hindlimb clasp, open field (HK MotorMonitor), Morris water maze, NOR, SOR).
- Conducted hemi-brain collections at a 30-day post-final-injection to assess pathology beyond the acute inflammatory period; performed fresh tissue microdissections of hippocampus and cortex and PFA-fixation of the contralateral hemisphere for histology, with subsequent tissue processing and analysis at MSU-GRRC.
- Supervised and trained undergraduate researchers in behavioral testing, animal handling, and tissue processing procedures.

2018 **Visiting Graduate Researcher**

Laboratory of Dr. David Morgan and Dr. Marcia Gordon | Michigan State University, Translational Science and Molecular Medicine (TSMM), Grand Rapids, MI

- Performed AAV-mediated stereotaxic injections of human tau constructs into bilateral hippocampus and anterior cortex (4 injection sites per animal, 25+ mice) for a tau seeding and propagation dose-optimization study; prepared viral vector and loaded Hamilton syringes for stereotaxic delivery.
- Conducted systematic post-operative monitoring and recovery scoring over 7 days (hypoactivity, hunched posture, social behavior, wound healing, body weight) with supportive care (mash diet) through recovery; performed

transcardial perfusions (4% PFA), microtome sectioning, and DAB IHC (HT7, AT8, pSer396).

2015–2020 **Graduate Researcher**

Laboratory of Dr. Yannick Marchalant, Neuroinflammation, Aging, and Alzheimer's Lab | Central Michigan University, Mount Pleasant, MI

- Conducted independent master's thesis research investigating prenatal inflammatory effects on amyloid burden and microglial density in the 5xFAD transgenic mouse model; managed colony maintenance, breeding, and genotyping (tail biopsy, multiplex PCR), and performed transcardial perfusions (4% PFA), vibratome sectioning, multi-label immunofluorescence staining (6E10, Iba1, Hoechst), widefield (ZEISS Axio Imager M1) and confocal (Nikon A1R) microscopy, and unbiased stereological quantification (MBF Stereo Investigator); in a parallel experimental arm, executed a single-dose systemic LPS challenge (1 mg/kg IP) in 2-month-old 5xFAD mice, with tissue collection at 4 months to examine effects of peripheral inflammation on amyloid pathology at the prodromal stage.

- Demonstrated that prenatal inflammatory exposure paradoxically reduced hippocampal amyloid plaque burden (6E10) by 46% (male) and 59% (female) in 5xFAD offspring of transgenic mothers relative to offspring of non-Tg mothers; maternal genotype did not alter microglial density, suggesting altered amyloid pathology independent of microglial quantity, highlighting early-life immune signaling as a potential modifiable contributor to amyloid pathology.

- Independently executed two concurrent preclinical studies (120+ animals) in the laboratory of Dr. Kevin Park utilizing inducible neuronal cell-cycle re-entry (NCCR) models of sporadic Alzheimer's disease: (1) TAg/OFF/MAPT triple-transgenic mice to investigate whether NCCR combined with humanized tau drives tau pathology and neuroinflammation; and (2) systemic PAMP administration in NCCR mice to examine whether peripheral immune activation exacerbates AD pathology and neuroinflammation, with PAMP-treated mice showing p-tau immunoreactivity (CP13; pS202) and glial reactivity (Iba1, CD45, GFAP) relative to controls; additionally, co-managed a PLX5622-mediated CSF1R microglial depletion study, observing a pronounced reduction of Iba1-positive microglia.

- Contributed to collaborative studies in cannabinoid pharmacology (Bruins et al.), demonstrating a biphasic effect of WIN 55,212-2 on mesenchymal stem cell (MSC) proliferation (peak at 1 μM), with MSC identity confirmed by FACS (Sca-1+/CD11b−); dendritic spine morphology in NL-G-F knock-in mice (Guentert et al.) using GFP-labeled neurons, confocal microscopy (Nikon A1R), and 3D reconstruction (MBF Neurolucida); and LRRK1/LRRK2 Parkinson's disease histology (Altemus et al.; Crombez et al.) via immunofluorescent and widefield microscopy (ZEISS Axio Imager M1); research presented across 10+ regional and national scientific meetings (2016–2019).

- Conducted work investigating ghrelin receptor (GHSR-1A) immunoreactivity in aged rats (15–24 months) via transcardial perfusion, vibratome sectioning, and immunofluorescence staining, prior to transitioning to master's thesis research.
- Supervised, trained, and mentored 30+ undergraduate and medical student researchers in transcardial perfusions, vibratome sectioning, immunohistochemistry, cell culture, behavioral testing, microscopy, and statistical analysis; designed and assigned independent research projects to advanced trainees, resulting in 6 undergraduate poster presentations at regional scientific meetings.

2013–2015 **Lead Psychometrist**

Clinical Director Dr. Gerard Williams, Delta Family Clinic South P.C., Flint, MI

- Administered, scored, and synthesized neuropsychological assessment batteries across pediatric and adult populations, including the Meyers Neuropsychological Battery (MNB), Wechsler scales (WPPSI, WISC, WAIS), MMPI-2, PAI, CARS-2, and GADS; evaluated neurocognitive and neurobehavioral profiles across clinical populations, including traumatic brain injury (TBI), stroke, mild cognitive impairment (MCI)/dementia, attention-deficit/hyperactivity disorder (ADHD), and autism spectrum disorder (ASD).
- Trained and supervised fellow psychometrists in standardized test administration and scoring protocols.

POSTER PRESENTATIONS

2023 **Keene, J.**, Desjarlais, T., Jackman, B., Srinageshwar, B., Finneran, D., Suresh, S., Henry, L., Gordon, M., Rossignol, J., Morgan, D., & Dunbar, G. (2023). "Inflammatory Stress Contributes to Tau Pathology in the PS19 Mouse Model of Alzheimer's Disease." Presented at the Michigan Society for Neuroscience (MISfN) Annual Meeting, Michigan State University (MSU), East Lansing, MI.

2022 **Keene, J**., Srinageshwar, B., Suresh, S., Stanley, R., Rossignol, J., & Dunbar, G. (2022). "Peripheral Inflammation and Physiological Hallmarks of Alzheimer's Disease." Presented at the Student Creative and Research Endeavors Exhibition (SCREE) Conference, CMU, Mount Pleasant, MI.

2019 Barrett, T**.**, **Keene, J**., Saito, T., Saido, T., & Park, K.H. (2019). "SV40T-Mediated Neuronal Dysregulation Induces NFT Pathology in APPNLF/NLFKI Mice." Presented at the Michigan Alzheimer's Disease Research Center (MADRC) 3rd Annual Research Symposium, Grand Rapids, MI and Michigan Physiological Society 6th Annual Meeting, Mount Pleasant, MI.

Keene, J., Madsen, L., Ginster, L., Bruins, A., Sawaya III, G., Barrett, T., Guentert, L., & Marchalant, Y. (2019). "Effects of Inflammatory Stress During the Asymptomatic Phase of Alzheimer's Disease." Presented at the MISfN Annual Meeting, Michigan State University (MSU), East Lansing, MI.

Bruins, A., **Keene, J**., & Marchalant, Y. (2019). “Synthetic Cannabinoid Agonist WIN 55,212-2 Modulates Proliferation and Migration of Bone Derived Murine Mesenchymal Stem Cells in Vitro.” Presented at the MISfN Annual Meeting, MSU, East Lansing, MI.

Madsen, L., Guentert, L., **Keene, J**., Dvorak, R., Shimer, F., Bruins, A., Sawaya III, G., & Marchalant, Y. (2019). “The Role of Inflammation and Sex on the Onset and Progression of Alzheimer's Disease.” Presented at SCREE, CMU, Mount Pleasant, MI.

MacLean, S., Barrett, T., Gallien, J., **Keene, J**., Brown, R., McCloskey, K., & Park, K. (2019). “The Effect of CSF1R Antagonist-mediated Ablation of Microglia in TAg/OFF Mice.” Presented at SCREE, CMU, Mount Pleasant, MI.

2018

Keene, J., Madsen, L., Ginster, L., Kunz, A., Bruins, A., Sawaya III, G., Guentert, L., Ridgell, R., & Marchalant, Y. (2018). “Role of Early Exposure to Inflammatory Stress on Plaque Deposition and Microglia Density in Alzheimer’s Disease.” Presented at the MISfN Annual Meeting, Wayne State University (WSU), Detroit, MI and SCREE, CMU, Mount Pleasant, MI.

Keene, J., Madsen, L., Ginster, L., Kunz, A., Bruins, A., Sawaya III, G., & Marchalant, Y. (2018). “Role of Prenatal Exposure to Inflammatory Stress on Microglia Density and Amyloidogenesis.” Presented at SCREE, CMU, Mount Pleasant, MI.

Madsen, L., **Keene, J**., Guentert, L., Bruins, A., Kunz, A., Ginster, L., Sawaya III, G., & Marchalant, Y. (2018). “Role of Inflammation and Sex in the Onset and Progression of Alzheimer’s Disease.” Presented at the MISfN Annual Meeting, WSU, Detroit, MI and SCREE, CMU, Mount Pleasant, MI.

Guentert, L., Madsen, L., **Keene, J**., & Marchalant, Y. (2018). “The Role of Inflammation and Sex on Dendritic Spines and Neuron Arborization in NL-G-F Mice.” Presented at the MISfN Annual Meeting, WSU, Detroit, MI and SCREE, CMU, Mount Pleasant, MI.

2017

Keene, J., Madsen, L., Pryde, C., Guentert, L., Crombez, E., McCrary, B., Cersosimo, L., Bailey, P., Kunz, A., Schwab, M., Gutierrez, N., Kemmerer, A., & Marchalant, Y. (2017). “Role of Neuroinflammation and Sex Differences in Alzheimer’s Disease.” Presented at the MISfN Annual Meeting, University of Michigan (U-M), Ann Arbor, MI and SCREE, CMU, Mount Pleasant, MI.

Madsen, L., **Keene, J.,** McCrary, B., & Marchalant, Y. (2017). “Role of Inflammation and Sex on the Onset and Progression of Alzheimer’s Disease.” Presented at the MISfN Annual Meeting, U-M, Ann Arbor, MI and SCREE, CMU, Mount Pleasant, MI.

Crombez, E., **Keene, J**., Altemus, M., Madsen, L., Marchalant, Y., & Johansen, J. (2017). “Influence of Neuroinflammation in the LRRK2 Mouse Model of Parkinson’s Disease.” Presented at SCREE, CMU, Mount Pleasant, MI.

2016 Crombez, E., **Keene, J**., Altemus, M., Madsen, L., Johansen, J., & Marchalant, Y. (2016). "Influence of Neuroinflammation in LRRK2 Mouse Model of Parkinson's Disease." Presented at the West Michigan Regional Undergraduate Science (WMRUGS) Research Conference, Grand Rapids, MI and Midwest/Great Lakes Undergraduate Research Symposium (mGluRs) in Neuroscience, Ohio State University, Columbus, Ohio.

Keene, J., & Marchalant, Y. (2016). "Ghrelin's Influence on Neurogenesis and Neuroinflammation in Aged Rats." Presented at the MISfN Annual Meeting, MSU, East Lansing, MI and BRAIN Center Symposium, CMU, Mount Pleasant, MI.

Keene, J., Antcliff, A., Crombez, E., Rossignol, J., & Marchalant, Y. (2016). "Cannabinoid Agonist WIN 55,212-2 Modulation of Bone Derived Mesenchymal Stem Cells." Presented at the MISfN Annual Meeting, MSU, East Lansing, MI.

Keene, J., & Marchalant, Y. (2016). "Ghrelin Induced Hippocampal Neurogenesis and Anti-inflammatory Effects in Aged Rats." Presented at SCREE, CMU, Mount Pleasant, MI.

GRANTS & AWARDS

2023 Field Neuroscience Chair Scholarship ($1,700) | Central Michigan University, Mount Pleasant, MI.

2021 Graduate Student Research and Creative Endeavors Grant ($800) | Central Michigan University, Mount Pleasant, MI. "*The Relationship Between Inflammation and the Pathological Hallmarks of Alzheimer's Disease*."

2020 Graduate Student Research and Creative Endeavors Grant ($800) | Central Michigan University, Mount Pleasant, MI. "*Examining the Role of Neuronal Cell Cycle Activation and Chronic Inflammation on Alzheimer's Disease-Related Molecular Pathways*."

2019 James and Catherine Steinmetz Graduate Scholarship ($1,000) | Central Michigan University, Mount Pleasant, MI.

2018 Graduate Student Research and Creative Endeavors Grant ($800) | Central Michigan University, Mount Pleasant, MI *"Effects of Inflammatory Stress During the Asymptomatic Phase of Alzheimer's Disease."*

James and Catherine Steinmetz Graduate Scholarship ($500) | Central Michigan University, Mount Pleasant, MI.

2016 James and Catherine Steinmetz Graduate Scholarship ($1,500) | Central Michigan University, Mount Pleasant, MI.

Graduate Student Research and Creative Endeavors Grant ($800) | Central Michigan University, Mount Pleasant, MI *"Ghrelin Induced Neurogenesis and Cell Survival in Aged Rats."*

2015–2016 Master's Fellowship in Experimental Psychology | Central Michigan University, Mount Pleasant, MI.

TEACHING EXPERIENCE

2018 Instructor: *Stress*. Department of Psychology, Central Michigan University, Mount Pleasant, MI.

Instructor: *Research Methods*, online. Department of Psychology, Central Michigan University, Mount Pleasant, MI.

Guest Lecturer: "Emotional and Motivated Behavior" and "Sleep and Dreaming" in course *Physiological Psychology*, Department of Psychology, Central Michigan University, Mount Pleasant, MI.

2017 Guest Lecturer: "Emotional Behaviors" in course *Behavioral Neuroscience*, Department of Psychology, Central Michigan University, Mount Pleasant, MI.

Teaching Assistant: *Developmental Psychology*, Department of Psychology, Central Michigan University, Mount Pleasant, MI. Guest Lecturer: "Middle Adulthood" and "Death and Afterlife Beliefs."

Teaching Assistant: *Introduction to Psychology*, Department of Psychology, Central Michigan University, Mount Pleasant, MI. Guest Lecturer: "Stress and Health."

2016 Guest Lecturer: "Internal Regulation" in course *Behavioral Neuroscience*, Department of Psychology, Central Michigan University, Mount Pleasant, MI.

Teaching Assistant: *Developmental Psychology*, Department of Psychology, Central Michigan University, Mount Pleasant, MI. Guest Lecturer: "Late Adulthood."

PRESENTATIONS, PANELS & WORKSHOPS

2021 Presenter: "The Role of Multiple Infection Mimetics on Tau Pathology and Neuroinflammation in PS19 Mice." Grand Rounds Lecture, Field Neurosciences Institute, Ascension St. Mary's, Saginaw, MI.

2018 Presenter: "Effects of Inflammatory Stress During the Asymptomatic Phase of Alzheimer's Disease." Experimental Psychology Seminar, Psychology Department, Central Michigan University, Mount Pleasant, MI.

Participant: Graduate Student Teaching Preparation Workshop, Psychology Department, Central Michigan University, Mount Pleasant, MI.

2017 Panelist: "Graduate Student Perspectives on Neuroscience Research and Graduate School Success." Neuroscience Night, Nu Rho Psi, Central Michigan University, Mount Pleasant, MI.

Presenter: "Sex Differences in Alzheimer's Disease using the 5xFAD Mouse Model." Experimental Psychology Department, Central Michigan University, Mount Pleasant, MI.

Presenter: "The Role of Neuroinflammation and Sex Differences in Alzheimer's Disease." Experimental Psychology Department, Central Michigan University, Mount Pleasant, MI.

Panelist: "Graduate School Application and Success Advice." Psi Chi Undergraduate Panel, Central Michigan University, Mount Pleasant, MI.

2016 Presenter: "Ghrelin's Influence on Neurogenesis and Neuroinflammation in Aged Rats." Experimental Psychology Department, Central Michigan University, Mount Pleasant, MI.

Panelist: "Alzheimer's Disease and Neuroinflammation." Brain Awareness Conference, Central Michigan University, Mount Pleasant, MI.

PROFESSIONAL SERVICE

2024 **Keene, J.** & Sanders, G. (**Eds**.). (2024). *Foundations of Neuropsychology Vol. 1*. OER Commons / Allan Hancock College, Santa Maria, CA. (Authored by Gallien, J.)

2019 **Ad Hoc Reviewer**, Archives of Clinical Neuropsychology (1 manuscript)

OUTREACH & SERVICE

2024–2025 Literacy Volunteer, Reading is Fundamental, Pleasant View Magnet School, Lansing School District, Lansing, MI. Classroom reading sessions (Pre-K, Kindergarten); book and educational materials distribution.

ORGANIZATIONS & MEMBERSHIPS

2017–2018 Graduate Committee member, Central Michigan University, Mount Pleasant, MI.

2017 Graduate Research and Creative Endeavors Grant Reviewer, Central Michigan University, Mount Pleasant, MI.

2016–2018 Experimental Psychology Organization Treasurer, Department of Psychology. Central Michigan University, Mount Pleasant, MI.

2016–2024 Michigan Society for Neuroscience member.

2015–2018 Experimental Psychology Organization member, Central Michigan University, Mount Pleasant, MI.

TECHNICAL SKILLS

Animal Models:

- **Neurodegenerative Disease**: Alzheimer's (transgenic: 3xTg, 5xFAD, PS19; knock-in: APP-KI [NL-F, NL-G-F], MAPT-KI) and Parkinson's (LRRK1/LRRK2 knock-out).
- **Inducible Systems**: Doxycycline-mediated neuronal cell cycle re-entry (TAg/OFF).

- **Viral Vector-Based**: AAV-mediated human tau seeding (bilateral hippocampus and anterior cortex).

In Vivo & Surgical Procedures:

- **Behavioral Testing:** Open field, Morris water maze (MWM), elevated plus maze (EPM), novel object recognition (NOR), spatial object recognition (SOR), Y-maze, hindlimb clasping, behavioral tracking and analysis software (HK MotorMonitor, EthoVision XT, ANY-maze).

- **Surgical Procedures:** Anesthesia (isoflurane, pentobarbital), stereotaxic survival surgery (cannula implantation, osmotic pump implantation, AAV-mediated injection), intraperitoneal (IP) injection, blood collection and serum fractionation, rodent handling and husbandry (mouse/rat), transcardial perfusion (4% PFA), brain microdissection (cortex and hippocampus), pharmacological microglial depletion (PLX5622/CSF1R inhibition).

Histology & Tissue Processing: Tissue sectioning (vibratome, microtome, cryostat), tissue cryopreservation, antigen retrieval (heat-induced/citrate, formic acid, enzymatic/trypsin), immunohistochemistry (DAB, IF, multi-label co-staining), Nissl staining, lipophilic neuronal tracing (DiI).

Microscopy, Stereology & Image Analysis:

- **Microscopy Platforms**: Confocal microscopy (Nikon A1R), widefield microscopy (brightfield and fluorescence; ZEISS Axio Scan.Z1, ZEISS Axio Imager M1).

- **Stereology & Morphometry**: Unbiased stereology (MBF Stereo Investigator: Optical Fractionator, Area Fraction Fractionator, Cavalieri method), dendritic morphometry and 3D reconstruction (MBF Neurolucida).

- **Image Analysis Software:** ImageJ, FIJI, NearCYTE.

Molecular Biology: PCR (nested and multiplex; including primer design), qRT-PCR (multiplex; TaqMan-based gene expression quantification), Western blot, ELISA (sandwich).

Cell Culture & In Vitro:

- **Culture Systems:** Primary cell culture (NSCs, MSCs, microglia, neurons) and established cell lines (U87 glioblastoma); cell passaging and cryopreservation.
- **Functional Assays:** Drug dose-response, cell proliferation, cell migration, scratch/wound healing.
- **Cell Analysis:** Hemocytometer/manual counting; FACS; ICC (live cell and post-fixed staining; targets: CB1, Iba1, Nestin, SOX2, CD73, CD90, CD105).
- **C. elegans Techniques:** Forward genetic screens, PCR, RNAi.

Quantitative Analysis & Biostatistics:

- **Software:** R programming, GraphPad Prism, SPSS.

- **Methods:** Experimental design, statistical modeling (ANOVA, Mann-Whitney U, Brunner-Munzel, Aligned Rank Transform, Robust Two-Way ANOVA, permutation, Spearman correlation), normality testing (Shapiro-Wilk, Anderson-Darling), post-hoc corrections (Tukey's, Holm); effect size measures (Partial Eta/Omega Squared, Cliff's Delta, Probability Index, Hodges-Lehmann), and data visualization.

Digital Communication: Scientific figure preparation (InkScape); portfolio website development (GitHub Pages).